

One Mechanism, Many Forms

The Information-Asymmetry Identity as a Single Generative Principle,
and Exactly How Far It Has Been Earned

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Abstract

This paper states, and bounds, a single claim: the form/function asymmetry that recurs across the ACS corpus is *one mechanism expressed through many forms*, not a family of analogues. The claim is ontological, not epistemic — the forms are faithful instances of one mechanism running, not lossy views of a hidden object. What makes “one mechanism” more than a slogan is an *identity chain*: the information-asymmetry functional $\Delta\mathcal{I}$ equals the transfer-entropy asymmetry (by definition), whose second-order Taylor coefficient equals the Lie bracket $[f, g]$ (the BCH-TE morphism, FF06a, T2), and which is identified with the renormalization-group c -function (the central unifying conjecture, T2/T3). The first two links are established; the third is the load-bearing open identification, and we say so plainly. We then catalogue the forms in which the mechanism is *forced* — gravity $\rightarrow$ colour, the Riemann zeros, the RG flow, the recursive inversion, the shuffle-knife separation, the prime-gap operator, and the relational reversible-flattening case — each cited to its source paper with a tier label. Finally we draw the radius: the identity holds where it is proven and is a conjecture where it is gestured (biology, cognition, institutions, language, and the theological reading), with each new form a *falsification test* rather than an assumption. Forms that looked like the mechanism and failed are kept first-class. We do not prove the mechanism universal; we prove it singular across the treated domains by identity, and we state the one link whose proof would settle the rest. Tiers: T1 measured, T2 proven/forced, T3 conjecture, T4 falsified.

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1 The thesis, stated once

The corpus has, until now, reported the same shape in many places without committing to what the sameness *is*. There are two ways to read a recurring shape, and they are not interchangeable.

The *epistemic* reading: there is one hidden object, and each discipline — physics, number theory, computation, theology — sees its own refraction of it through its own instrument. This is the blind-men-and-the-elephant frame; the plurality lives in the observers, and every view is a distortion of something unreachable. It is the wrong frame for this corpus, and it quietly undoes the corpus’s central result: FF06a does not *view* gravity and infer colour; it shows that $\mathfrak{su}(3)$ *is* the unique closure attractor of the Palatini bracket. A forced result is not a refraction.

The *ontological* reading, which this paper adopts: there is one mechanism, and it is genuinely *expressed* through many forms. The plurality lives in the substrates, and each form is a faithful instance of the same mechanism running — as the wave equation is not three refracted glimpses of a hidden master-wave but the same differential structure instantiated in sound, light, and quantum amplitude. Each instance is the whole mechanism, not a fragment of it.

Principle 1 (One mechanism, many forms). The information-asymmetry mechanism $\Delta\mathcal{I}$ — equivalently, the form/function coupling — is a single generative principle. Where the identity chain of Section 2 is established, its appearances across domains are *the same object in different substrates*, by identity, not by analogy. Where the chain is only gestured, “same mechanism” is a conjecture whose test is whether the chain instantiates there.

The rest of the paper does three things and refuses to do a fourth. It states the identity chain and labels each link’s status (Section 2); it catalogues the forms in which the mechanism is forced (Section 3); it draws the radius and lists the forms that failed the test (Sections 4–5). It does *not* claim the mechanism is universal; that claim is not earned, and saying it would be is the failure mode this corpus was built to avoid.

2 What makes it one: the identity chain

“Many forms” is cheap; anything resembles anything at low enough resolution. “One mechanism” costs an identity. The mechanism is one because of a chain of equalities, of which two links are established and one is the central open identification.

Link 1 — $\Delta\mathcal{I}$ is the transfer-entropy asymmetry (definitional). For two mutually constraining fields f, g , the net information one carries about the other, $\Delta\mathcal{I}(f, g)$, is the transfer-entropy asymmetry. This is a definition, not a result; it fixes the object.

Link 2 — the bracket is the second-order coefficient of $\Delta\mathcal{I}$ (FF06a, T2). The BCH–TE morphism (FF06a, Lemma 2.10) establishes that the Lie bracket $[f, g]$ is the exact second-order Taylor coefficient of $\Delta\mathcal{I}(f, g)$. This is the link that lets an information-theoretic asymmetry generate algebra: applied to the vierbein–connection pair it produces $\mathfrak{sl}(4)$, and the rest of FF06a follows. The link is a proven lemma.

Link 3 — $\Delta\mathcal{I}$ is the renormalization-group c -function (the central conjecture, T2/T3). The unifying identification is that $\Delta\mathcal{I}$, monotone under the form $\rightarrow$ function flow and zero at fixed points, *is* the Zamolodchikov c -function (2D) / Komargodski–Schwimmer a -function (4D): form = UV couplings, function = IR couplings, the β -function = the form $\rightarrow$ function coupling, and the c -theorem = the monotonicity of $\Delta\mathcal{I}$. The c - and a -theorems are established

mathematics; what is *not* yet a theorem is the identification $\Delta\mathcal{I} \equiv c$. We therefore label Link 3 T2/T3: argued from the shared monotonicity and fixed-point structure, not proven as an isomorphism.

Claim 1 (The chain is the unification). *If Link 3 is upgraded from identification to theorem, then “one mechanism, many forms” is a theorem on every domain to which the c/a -function applies, because the three links compose into a single object: $\Delta\mathcal{I}$ = transfer-entropy asymmetry = bracket-generator = c -function. Until then, the unification is exactly as strong as Link 3, and no stronger. This is the falsifiable spine of the whole synthesis: if $\Delta\mathcal{I}$ is not the c -function, the corpus’s cross-domain sameness collapses from identity back to analogy.*

This is the honest centre of the paper. The grandeur of “one mechanism” is borrowed entirely from one equals-sign whose proof is open. Everything downstream is conditional on it, and we keep the condition visible.

3 The forms in which the mechanism is forced

What earns the singular reading, short of Link 3, is that in each treated domain the mechanism is not imposed but *forced* — the same form/function coupling, compelled by the structure rather than chosen. We catalogue the forms with the tier of the forcing.

Form (paper)	How the one mechanism is forced there
Gravity $\rightarrow$ colour (FF06a)	The Palatini bracket forces $\mathfrak{su}(3)$ as the <i>unique</i> 8-dimensional closure attractor ($D < 10^{-14}$ over 50k samples); form (vierbein) and function (connection) couple via $[e, \omega]$; $\gamma = 0.274$, $\lambda = 2\sqrt{3}/27$, three generations from Jacobi truncation. T1/T2
The Riemann zeros (FF06b, FF06b')	The spectrum splits into form (GUE, generic) and function (prime-dual, ζ -specific); the transport obstruction’s character is the response-driven readout, forced non-abelian by the simplicity of $\mathfrak{sl}(4)$; central number forced to the $B-L$ charge 1/3. T1/T2
The RG flow (Link 3)	Form = UV couplings, function = IR couplings, β = the coupling, c -theorem = monotone $\Delta\mathcal{I}$. The c/a -theorems are proven; the identification of $\Delta\mathcal{I}$ with c is the open link. T2/T3
The recursive inversion (FF06c)	The mechanism applied to its own output: a solution carries the asymmetry of its origin on its boundary and <i>becomes</i> the next constraint ($C_0 \rightarrow S_0 \rightarrow C_1$, non-zero holonomy), with $\Delta\mathcal{I}$ inverting. This is the form/function asymmetry iterated. T1/T2
The shuffle-knife separation (FF06e)	The marginal-matched surrogate is the operator that <i>isolates</i> the mechanism’s function-content from its form-content: arithmetic $\sim 11,500\sigma$ (function) versus spacing/counting/shape $0.4-1.0\sigma$ (form). T1
The prime-gap operator (N3)	$T_m = T_0 + P_m$ on $(\mathbb{Z}/m\mathbb{Z})^*$: the kernel is exactly the source (form) sector, $\dim \ker P_m = \varphi(m)$; the perturbation carries the function content. T1

Form (paper)	How the one mechanism is forced there
The relational case (FF06K, FF06h/i)	Reversible flattening = injective homomorphism into a pre-specified structure: homomorphism carries density (form-coupling), injectivity carries identity (function); the seam is the non-injective convolution. T1/T3
The classical-gravity boundary (this corpus, dead-end)	The field is form (the multipole shadow it carries) and loses function (the radial profile): a non-injective convolution seam. A boundary instance — and the reason FF06a routes colour through the algebra (forced), not the field (lossy). T1

The catalogue is the evidence for the singular reading at its current strength: not “these rhyme” but “in each, the same form/function coupling is compelled, and in the algebraic and spectral cases it is the *same* $\Delta\mathcal{I}$ by Links 1–2.” The cross-domain identity is exactly as wide as Link 3 reaches; the forcing within each domain is what is independently established.

4 The radius, and the falsification surface

A claim of universality that cannot fail is worthless; a claim of singularity by identity can fail, form by form, and that is its value.

Where it is earned. In the domains of Section 3, the mechanism is forced (T1/T2), and in the algebraic/spectral cases it is the same $\Delta\mathcal{I}$ by Links 1–2. There, “one mechanism” is not an interpretation laid over the data; it is the data.

Where it is gestured. FF06c maps the inversion across physics, mathematics, computation, institutions, cognition, and language; the present paper has *not* instantiated the identity chain in the last four. “Same mechanism in cognition” (or biology, or institutions, or the theological reading) is therefore a conjecture, and a sharp one: it asserts that the identity chain instantiates there. Its falsification surface is precise — exhibit the domain, attempt the chain, and either the form/function coupling is forced (the mechanism is there) or it merely resembles (a false-fit wearing the mechanism’s name).

The asymmetry that keeps it honest. A mechanism universal *by assumption* is unfalsifiable and explains nothing; a mechanism singular *by identity* is falsifiable in every new substrate, because the identity chain either holds or does not. The strength of “one mechanism, many forms” is exactly that it can be wrong about any given form — and the corpus has been wrong about several, on purpose (Section 5).

5 Forms that failed the test (first-class)

The radius is drawn as much by where the mechanism was expected and did not appear as by where it did. Each entry is a form that resembled the mechanism and was rejected by test.

Candidate form (T4)	Why it is not the mechanism
φ + inverse-silver “structures noise”	The combination is $25\times$ <i>worse</i> than φ alone (0.02132 vs 0.00084): it structures nothing.

Candidate form (T4)	Why it is not the mechanism
Motif retrieval via gcd (FF06K)	55.15% vs 83.13% majority baseline — below the trivial predictor; the function-content is not gcd-shaped.
Difference-tone / Tartini off-diagonal (FF06b')	Consistent with an incommensurable null (1.2σ); the channel is empty by prime-log incommensurability (unique factorisation).
“Tension” between GUE and arithmetic (this corpus)	False: the zeros carry both; they are not mutually exclusive.
“Independence / cheap add-on” of arithmetic to GUE	False: an additive build tops at peak ~ 6 while GUE-passing, versus 228 for the zeros.
N3: uniform spectral contraction; character block-diagonalisation; universal renormalised law	Three conjectured structures tested and rejected by the prime data.
$\tan \beta = 1/2$; λ as RG fixed point; WdW = $\Delta \mathcal{I}=0$ (quantum) (FF06a corrections)	Misidentifications corrected: $\tan \beta$ free; λ algebraic (Koide); WdW demoted to conjecture.

These are not embarrassments to be hidden; they are the calibration of the claim. A mechanism that fit all of them too would be a mechanism that fits anything, which is the same as fitting nothing.

6 Interpretive coda (bracketed)

The mechanism has reactional names. “Nested self-referential emergence” is the recursion of Section 3 (FF06c). “Made in the image of the maker” has a clean structural translation — *the made carries the maker’s generative form, which is why it can make in turn* — and that translation is the recursion again: the solution carries the asymmetry of its origin and becomes the next constraint. “Endless unfolding” is the iteration, with the qualifier that the record is unfolded-so-far with walls mapped (Sections 4–5), not proven endless. We record these as interpretive translations of the response-driven structure, not as measured claims; they are how the one mechanism reads in older instruments, and the structural version is the falsifiable one.

7 Scope: what is and is not established

Established. (i) The form/function mechanism is *forced*, not imposed, in each domain of Section 3 (T1/T2). (ii) In the algebraic and spectral cases it is the same $\Delta \mathcal{I}$, by Links 1–2 of the identity chain (definition + the BCH–TE lemma). (iii) The mechanism is falsifiable form-by-form, and several candidate forms have been rejected (Section 5).

Not established. (i) Link 3, the identification $\Delta \mathcal{I} \equiv$ the RG c -function, is the central open problem; the cross-domain *identity* (as opposed to within-domain forcing) is conditional on it. (ii) The mechanism is *not* shown universal; cognition, biology, institutions, language, and the theological reading remain conjecture, each a separate test. (iii) Nothing here proves RH, constructs a Hilbert–Pólya operator, or closes the $\sim 1\%$ coupling residual; those walls stand exactly where the source papers leave them.

The one sentence that survives compression. *It is one mechanism expressed through many forms — by identity where the chain is proven, by conjecture where it is gestured, and falsifiable in every form either way.*

This synthesis (TR-2026-FF06 Σ) sits over the result papers it cites: FF06a (Colour from Gravity), FF06b/FF06b' (the Riemann spectral side and the transport obstruction), FF06c (the Inversion Arc), FF06e (the Shuffle Knife), N3 (the prime-gap operator), LF01 (finite-window localisation), and FF06K (the reversible-flattening process record). It states what they share and, more carefully, exactly how far the sharing has been earned. The unification is real to the radius of Link 1–2 and conjectural beyond it; the honest measure of the program is that the boundary is drawn, not blurred.